Amortization, accommodation and approximation gaps

Draft of notation and Lemma 1

1 Setting and notation

Model. We consider a simple hierarchical model with decoder parameters θ and one local latent variable per datapoint,

$$p_\theta(z, x) = \prod_{n=1}^{N} p(z_n)\, p_\theta(x_n \mid z_n)$$

and data $x = (x_1, \ldots, x_N) \in \mathcal{X}$. The decoder is a point estimate (variational EM), for simplicity the prior on θ is flat.[1] We write $\log p_\theta(x) = \sum_n \log p_\theta(x_n)$.

Classes.

- Θ: the class of decoders (an architecture), the same for every method.
- $\mathcal{Q} = \{q(\cdot, \nu) : \nu \in U\}$: the variational family (e.g. FFG or a flow), with parameter space U.
- $\mathcal{F}$: the class of encoders, i.e. maps $f : \mathcal{X} \to U$. The family $\mathcal{Q}$ is implied by the codomain of $\mathcal{F}$, and F-VI optimises over all of U.

Objectives. The ELBO of datapoint n for parameters ν, decoder θ and a variational family $\mathcal{Q}$:

$$\mathcal{L}_n(\nu, \theta, \mathcal{Q}) = \mathbb{E}_{q(z,\nu)}\big[\log p(z) + \log p_\theta(x_n \mid z) - \log q(z, \nu, \mathcal{Q})\big] = \log p_\theta(x_n) - \mathrm{KL}\big(q(\cdot, \nu, \mathcal{Q}) \,\|\, p_\theta(\cdot \mid x_n)\big)$$

For a decoder θ and a variational family $\mathcal{Q}$, the best ELBO of F-VI and of A-VI are

$$\mathcal{L}_F(\theta, \mathcal{Q}) \coloneqq \sup_{\nu_1, \ldots, \nu_N \in U} \sum_{n=1}^{N} \mathcal{L}_n(\nu_n, \theta, \mathcal{Q}) \qquad \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q}) \coloneqq \sup_{f \in \mathcal{F}} \sum_{n=1}^{N} \mathcal{L}_n\big(f(x_n), \theta, \mathcal{Q}\big)$$

The per-datapoint suboptimality of ν is

$$\delta_n(\nu, \theta, \mathcal{Q}) \coloneqq \sup_{\nu' \in U} \mathcal{L}_n(\nu', \theta, \mathcal{Q}) - \mathcal{L}_n(\nu, \theta, \mathcal{Q}) \geq 0$$

Remark 1 (what separation needs). Two structural assumptions are used: each term of $\log p_\theta(z, x)$ involves a single z_n (simple hierarchical model), and q is a product over datapoints. In a hidden Markov model, $\log p(z_n \mid z_{n-1})$ couples z_n and z_{n-1}, so its expectation depends on (ν_n, ν_{n-1}): (**??**) fails and the optimal ν_n depends on its neighbours.

[1] With a prior $p(\theta)$, add $\log p(\theta)$ to $\log p_\theta(x)$, $\mathcal{L}_F(\theta)$ and $\mathcal{L}_A(\theta, \mathcal{F})$ below, nothing else changes.

Decoders. We assume the following maximisers exist in Θ and the maxima are finite:

$$\hat{\theta}_{\mathrm{ML}} \in \arg\max_{\theta\in\Theta} \log p_\theta(x) \qquad \hat{\theta}_F \in \arg\max_{\theta\in\Theta} \mathcal{L}_F(\theta, \mathcal{Q}) \qquad \hat{\theta}_A \in \arg\max_{\theta\in\Theta} \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q})$$

$\hat{\theta}_A$ is the decoder of a VAE trained to its optimum, $\hat{\theta}_F$ that of variational EM with free local parameters. If the maxima are not attained, the statements below hold up to ε with ε-maximisers. The maximisers over all decoders, $\theta^*_{\mathrm{ML}}, \theta^*_F, \theta^*_A$, are the special case where Θ is unconstrained, whenever they exist.

Definition 1 (Amortization gaps)**.** For a decoder $\theta \in \Theta$ (Θ respresents the decoder class) and a variational family $\mathcal{Q}$:

$$\mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q}) \coloneqq \mathcal{L}_F(\theta, \mathcal{Q}) - \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q})$$

$$\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) \coloneqq \sup_{\theta\in\Theta} \mathcal{L}_F(\theta, \mathcal{Q}) - \sup_{\theta\in\Theta} \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q}) = \mathcal{L}_F(\hat{\theta}_F, \mathcal{Q}) - \mathcal{L}_A(\hat{\theta}_A, \mathcal{F}, \mathcal{Q})$$

$\mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q})$ is the amortization gap at a fixed decoder, it is the quantity measured by Cremer et al. (2018) at $\theta = \hat{\theta}_A$. $\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q})$ compares the best F-VI and the best A-VI objectives, each with its own decoder, as described by Margossian and Blei (2024). Note that when $\Theta = \{\theta\}$ (ie the decoder is constrained) the above notations match: $\mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}) = \mathcal{G}_{\mathrm{am}}(\{\theta\}, \mathcal{F})$.

Lemma 1. *In a simple hierarchical model, for every class $\mathcal{F}$ and $\mathcal{Q}$:*

(i) (fixed decoder) *for every $\theta \in \Theta$,*

$$\mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q}) = \inf_{f\in\mathcal{F}} \sum_{n=1}^{N} \delta_n\big(f(x_n), \theta, \mathcal{Q}\big)$$

(ii) (decoder class)

$$\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) = \inf_{\theta\in\Theta} \Big[\big(\mathcal{L}_F(\hat{\theta}_F, \mathcal{Q}) - \mathcal{L}_F(\theta, \mathcal{Q})\big) + \mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q})\Big],$$

and the infimum is attained at $\theta = \hat{\theta}_A$.

Proof. (i) For $f \in \mathcal{F}$, by separation,

$$\mathcal{L}_F(\theta, \mathcal{Q}) - \sum_n \mathcal{L}_n(f(x_n), \theta, \mathcal{Q}) = \sum_n \Big[\sup_\nu \mathcal{L}_n(\nu, \theta, \mathcal{Q}) - \mathcal{L}_n(f(x_n), \theta, \mathcal{Q})\Big] = \sum_n \delta_n(f(x_n), \theta, \mathcal{Q})$$

Taking the supremum over f of $\sum_n \mathcal{L}_n(f(x_n), \theta)$ gives (i).

(ii) For every θ, $\mathcal{L}_A(\theta, \mathcal{F}) = \mathcal{L}_F(\theta) - \mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F})$, so

$$\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) = \mathcal{L}_F(\hat{\theta}_F, \mathcal{Q}) - \sup_\theta \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q}) = \inf_\theta \big[\mathcal{L}_F(\hat{\theta}_F, \mathcal{Q}) - \mathcal{L}_F(\theta, \mathcal{Q}) + \mathcal{G}_{\mathrm{am}}(\theta, \mathcal{F}, \mathcal{Q})\big]$$

At $\theta = \hat{\theta}_A$ the bracket equals $\mathcal{L}_F(\hat{\theta}_F) - \mathcal{L}_A(\hat{\theta}_A, \mathcal{F}) = \mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F})$. □

Corollary 2 (two inequalities)**.**

$$\mathcal{G}_{\mathrm{am}}(\hat{\theta}_A, \mathcal{F}, \mathcal{Q}) \le \mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) \le \mathcal{G}_{\mathrm{am}}(\hat{\theta}_F, \mathcal{F}, \mathcal{Q})$$

Proof. Expand the definitions: $\mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) - \mathcal{G}_{\mathrm{am}}(\hat{\theta}_A, \mathcal{F}, \mathcal{Q}) = \mathcal{L}_F(\hat{\theta}_F, \mathcal{Q}) - \mathcal{L}_F(\hat{\theta}_A, \mathcal{Q})$, nonnegative since $\hat{\theta}_F$ maximises $\mathcal{L}_F$, and $\mathcal{G}_{\mathrm{am}}(\hat{\theta}_F, \mathcal{F}, \mathcal{Q}) - \mathcal{G}_{\mathrm{am}}(\Theta, \mathcal{F}, \mathcal{Q}) = \mathcal{L}_A(\hat{\theta}_A, \mathcal{F}, \mathcal{Q}) - \mathcal{L}_A(\hat{\theta}_F, \mathcal{F}, \mathcal{Q})$, nonnegative since $\hat{\theta}_A$ maximises $\mathcal{L}_A$. □

Remark 2. The left inequality says that the amortization gap measured at the decoder of the trained VAE underestimates $\mathcal{G}_{\text{am}}(\Theta, \mathcal{F}, \mathcal{Q})$, by the loss of F-VI when it is restricted to that decoder. The right inequality is part (ii) of Lemma 1 evaluated at $\theta = \hat{\theta}_F$. With $\Theta = \{\theta\}$, all three quantities coincide.

Definition 2 (Total, accommodation approximation and amortization gaps)**.** For a given decoder $\theta \in \Theta$, we note

$$\mathcal{G}(\theta, \mathcal{F}, \mathcal{Q}) \coloneqq \log p_{\hat{\theta}_{\text{ML}}}(x) - \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q}) \qquad \mathcal{G}_{\text{ac}}(\theta) \coloneqq \log p_{\hat{\theta}_{\text{ML}}}(x) - \log p_\theta(x)$$

$$\mathcal{G}_{\text{ap}}(\theta, \mathcal{Q}) \coloneqq \log p_\theta(x) - \mathcal{L}_F(\theta, \mathcal{Q}) \quad \mathcal{G}_{am}(\theta, \mathcal{F}, \mathcal{Q}) = \mathcal{L}_F(\theta, \mathcal{Q}) - \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q})$$

$\mathcal{G}(\theta, \mathcal{F}, \mathcal{Q})$ is the log-likelihood that the trained VAE loses relative to the best decoder of the class. $\mathcal{G}_{\text{ac}}(\theta)$ is the log-likelihood that the decoder gives up by being trained with the ELBO and the encoder rather than by maximum likelihood (accommodation). $\mathcal{G}_{\text{ap}}(\theta, \mathcal{Q})$ is the approximation gap at decoder θ and $\mathcal{G}_{\text{am}}(\theta, \mathcal{F}, \mathcal{Q})$ is the amortization gap.

Proposition 3 (decomposition)**.**

$$\mathcal{G}(\theta, \mathcal{F}, \mathcal{Q}) = \mathcal{G}_{\text{ac}}(\theta) + \mathcal{G}_{\text{ap}}(\theta, \mathcal{Q}) + \mathcal{G}_{\text{am}}(\theta, \mathcal{F}, \mathcal{Q}),$$

and each term is nonnegative.

Proof. The identity telescopes: $\big[\log p_{\hat{\theta}_{\text{ML}}} - \log p_\theta\big] + \big[\log p_\theta - \mathcal{L}_F(\theta, \mathcal{Q})\big] + \big[\mathcal{L}_F(\theta, \mathcal{Q}) - \mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q})\big]$. $\mathcal{G}_{\text{ac}} \geq 0$ because $\hat{\theta}_{\text{ML}}$ maximises $\log p_\theta(x)$ over Θ, $\mathcal{G}_{\text{ap}} \geq 0$ because the ELBO is at most the log-likelihood, $\mathcal{G}_{\text{am}} \geq 0$ because every encoder output is an admissible value of ν_n. □

Remark 3 (reading). Since $\log p_{\hat{\theta}_{\text{ML}}}(x)$ does not depend on θ, training a VAE, i.e. maximising $\mathcal{L}_A(\theta, \mathcal{F}, \mathcal{Q})$ over θ, minimises the sum of the three terms evaluated at θ. The decoder may therefore trade likelihood ($\mathcal{G}_{\text{ac}}$) for smaller approximation and amortization gaps. Cremer et al. (2018) measure $\mathcal{G}_{\text{ap}}(\hat{\theta}_A)$ and $\mathcal{G}_{\text{am}}(\hat{\theta}_A, \mathcal{F})$, $\mathcal{G}_{\text{ac}}$ is not measured.